# Lecture 11: Node Voltage Analysis

## EE201 – Circuit Theory I

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current voltage, node voltage $e$ supernode

# Review: two ways to solve an LTI resistive circuit

## Transform the circuit

- Substitution, equivalents
- Series, parallel
- Source transformation

Many circuits, not all

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- KVL: $\mathbf{v} = A^T\mathbf{e}$
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Only KCL is left

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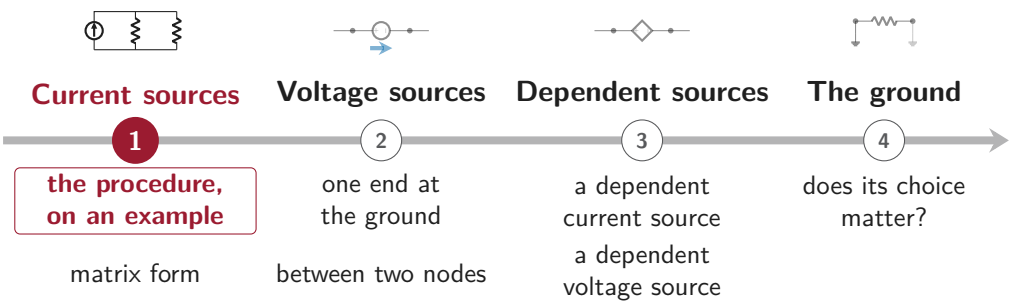
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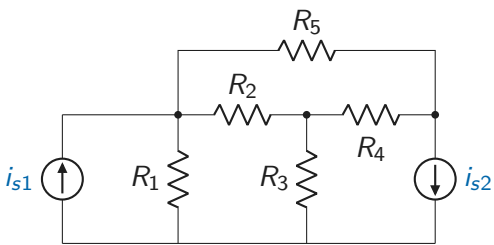
- Original circuit
- Unknowns: $n-1$ node voltages
- Then every $v$ and $i$
- Every LTI resistive circuit

# Roadmap


Current sources
Voltage sources
Dependent sources
The ground
1
2
3
4
the procedure, on an example
matrix form
one end at the ground
between two nodes
a dependent current source
a dependent voltage source
does its choice matter?

# Example 11.1: node voltage analysis step by step

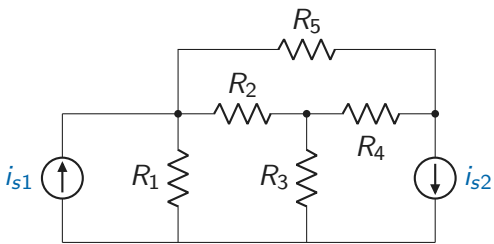
For now: only resistors and independent current sources




**Question.** How many unknowns?

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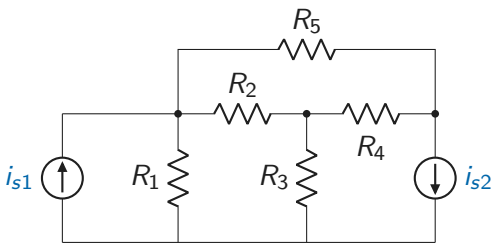



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$b = 7$: 14 unknowns

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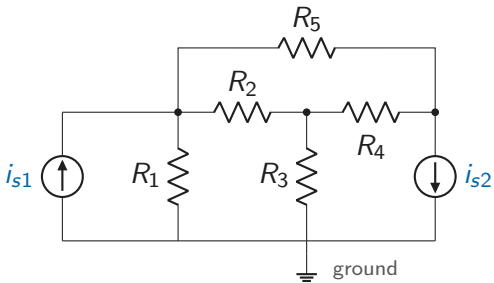



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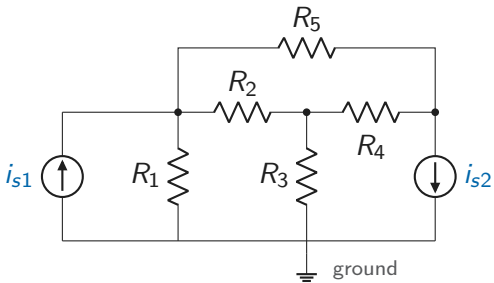
Node voltage analysis: 3

# Step 1: choose a reference node




Reference node: the ground

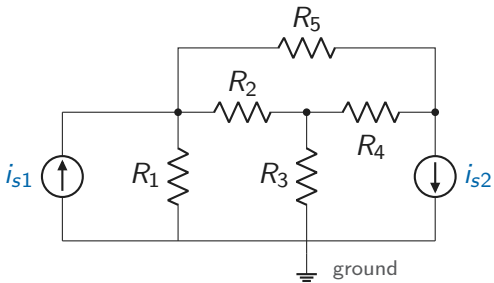
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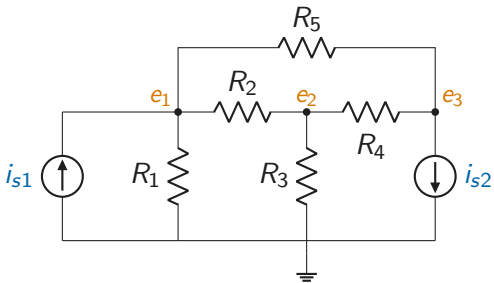



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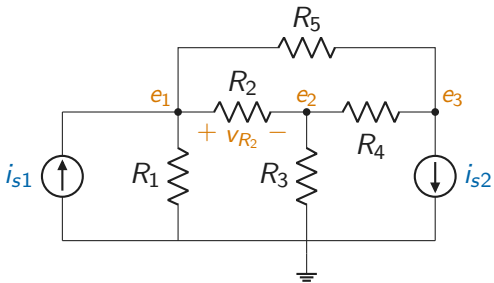
Any node can be the ground

# Step 2: label the node voltages




Unknowns: $e_1$, $e_2$, $e_3$

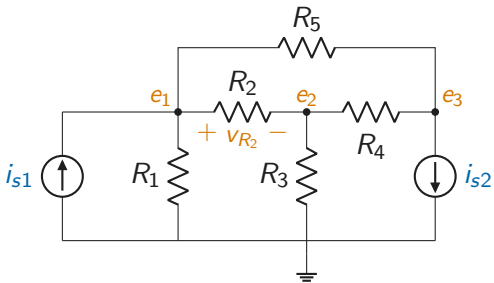
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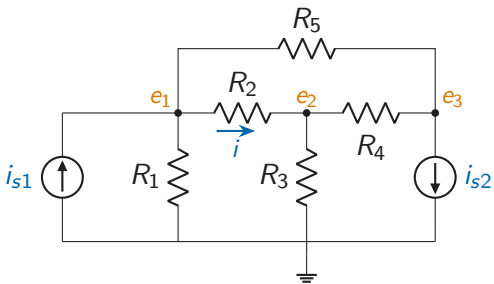
$v_{R_2} = e_1 - e_2$

KVL holds automatically:

$v_{R_1} = v_{R_2} + v_{R_3}$

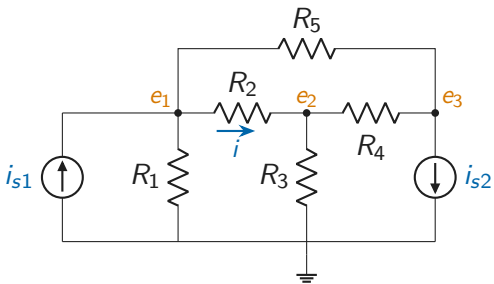
$e_1 = (e_1 - e_2) + e_2$

# Step 3: write the currents with node voltages




**Question.** $i$ leaving node 1 through $R_2$?

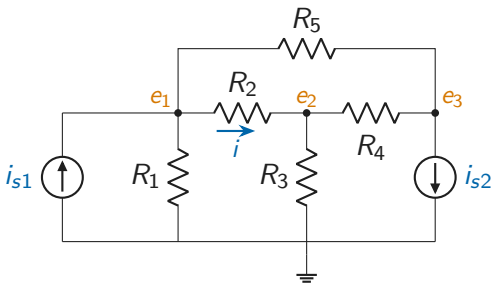
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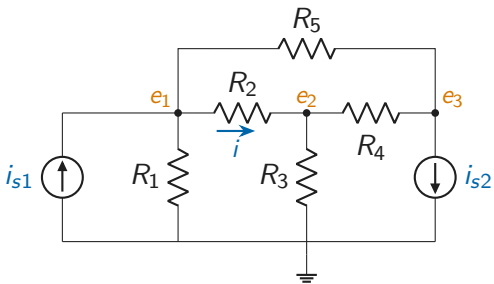



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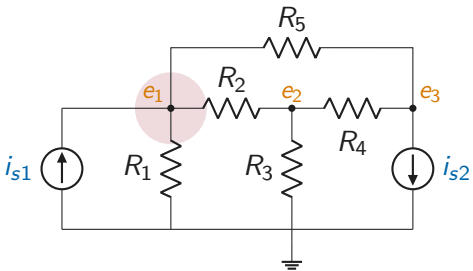

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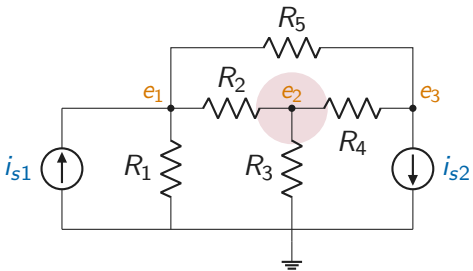
# Step 4: KCL at every node except the ground




**Node 1:**

$$-i_{s1} + \frac{e_1}{R_1} + \frac{e_1 - e_2}{R_2} + \frac{e_1 - e_3}{R_5} = 0$$

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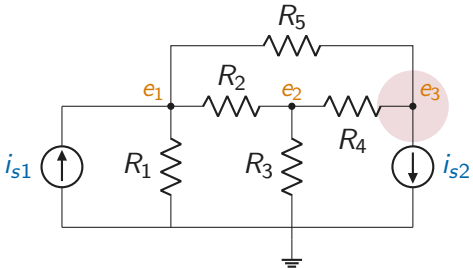

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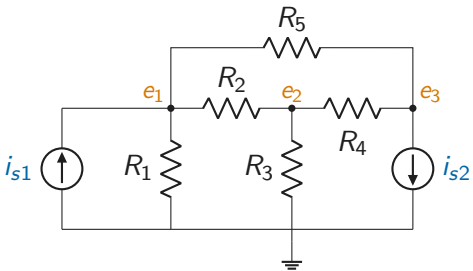
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**Node 3:**

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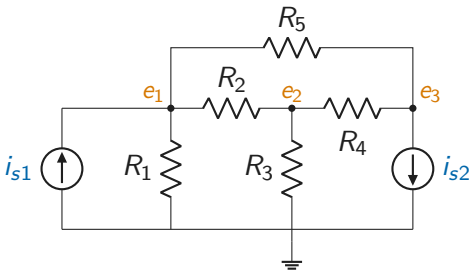
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Source into the node: minus

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Source into the node: minus

3 equations, 3 unknowns

# Roadmap









**Current sources** | **Voltage sources** | **Dependent sources** | **The ground**

1 — 2 — 3 — 4

the procedure, on an example ✓

**matrix form**

one end at the ground

between two nodes

a dependent current source

a dependent voltage source

does its choice matter?

# Example 11.1 in matrix form

$G_k = 1/R_k$:

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\begin{aligned}
(G_1 + G_2 + G_5)\, e_1 - G_2\, e_2 - G_5\, e_3 &= i_{s1} \\
-G_2\, e_1 + (G_2 + G_3 + G_4)\, e_2 - G_4\, e_3 &= 0 \\
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**Question.** One matrix equation?

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$Y_n$: node admittance matrix

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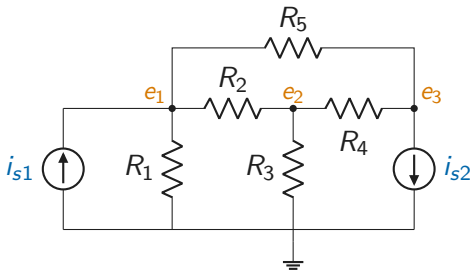
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$$

$Y_n$: node admittance matrix

$\mathbf{i}_n$: source currents entering each node

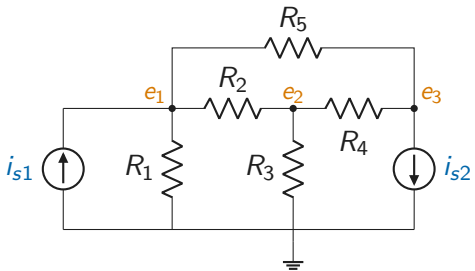
# Writing $Y_n$ directly from the circuit




$$\begin{bmatrix} G_1+G_2+G_5 & -G_2 & -G_5 \\ -G_2 & G_2+G_3+G_4 & -G_4 \\ -G_5 & -G_4 & G_4+G_5 \end{bmatrix}$$

**Question.** Read $Y_n$ from the circuit?

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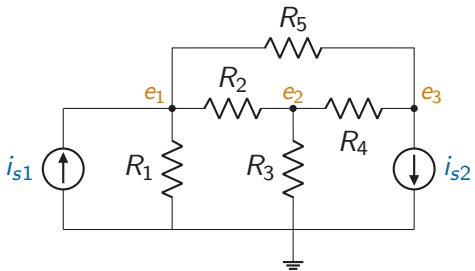



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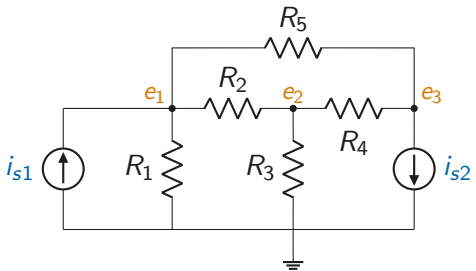

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**Off-diagonal** $(k,j)$:
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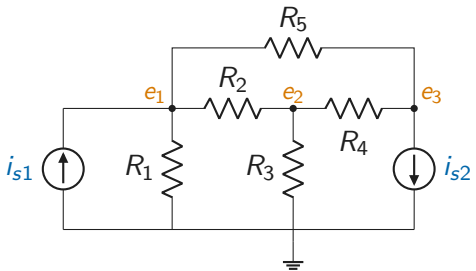
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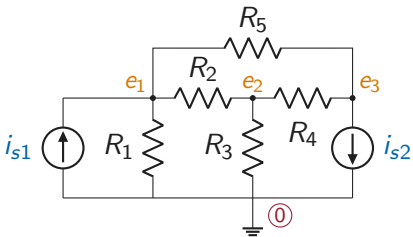
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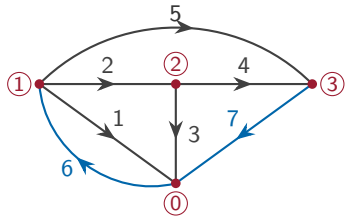
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$Y_n$ is symmetric

# $Y_n$ from the graph: the reduced incidence matrix

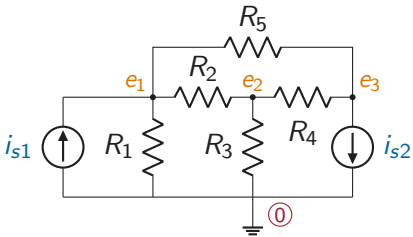


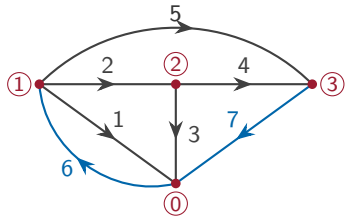




1–5: $R_1, \ldots, R_5$; 6, 7: $i_{s1}$, $i_{s2}$; ⓪: ground

**Question.** Reduced incidence matrix $A$?

# $Y_n$ from the graph: the reduced incidence matrix




1–5: $R_1, \ldots, R_5$; 6, 7: $i_{s1}$, $i_{s2}$; ⓪: ground

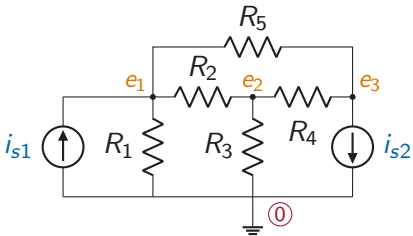
**Question.** Reduced incidence matrix $A$?

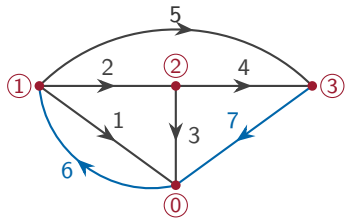
One row per non-ground node

$+1$ leaves, $-1$ enters

$$A = \begin{array}{c} \\ ① \\ ② \\ ③ \end{array} \begin{array}{c} \begin{array}{ccccccc} 1 & 2 & 3 & 4 & 5 & 6 & 7 \end{array} \\ \left[\begin{array}{ccccccc} & & & & & & \end{array}\right] \end{array}$$

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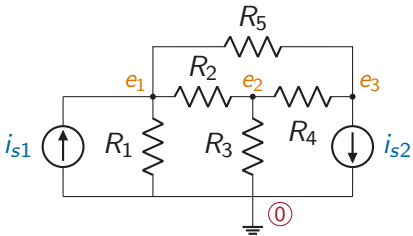
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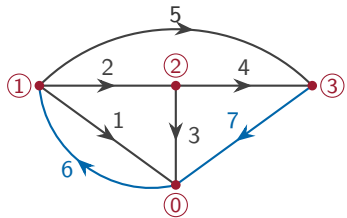
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$$A = \begin{matrix} \\ ① \\ ② \\ ③ \end{matrix} \begin{matrix} \begin{matrix} 1 & 2 & 3 & 4 & 5 & 6 & 7 \end{matrix} \\ \begin{bmatrix} 1 & 1 & 0 & 0 & 1 & -1 & 0 \\ & & & & & & \\ & & & & & & \end{bmatrix} \end{matrix}$$

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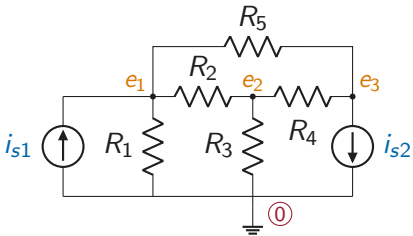
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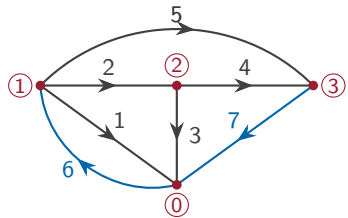
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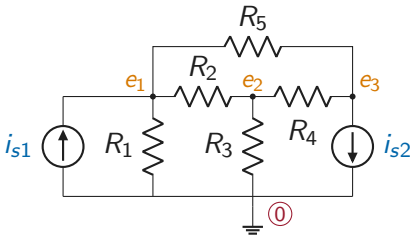
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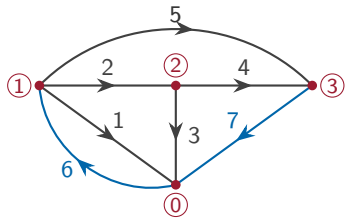
One row per non-ground node

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$$A = \begin{array}{c} \\ ① \\ ② \\ ③ \end{array} \begin{array}{c} \begin{array}{ccccccc} 1 & 2 & 3 & 4 & 5 & 6 & 7 \end{array} \\ \begin{bmatrix} 1 & 1 & 0 & 0 & 1 & -1 & 0 \\ 0 & -1 & 1 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & -1 & -1 & 0 & 1 \end{bmatrix} \end{array}$$

# $Y_n$ from the graph: the reduced incidence matrix




1–5: $R_1, \ldots, R_5$; 6, 7: $i_{s1}$, $i_{s2}$; ⓪: ground

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One row per non-ground node

$+1$ leaves, $-1$ enters

$$A = \begin{array}{c} \\ ① \\ ② \\ ③ \end{array} \overset{\begin{array}{ccccccc} 1 & 2 & 3 & 4 & 5 & 6 & 7 \end{array}}{\left[\begin{array}{ccccc|cc} 1 & 1 & 0 & 0 & 1 & -1 & 0 \\ 0 & -1 & 1 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & -1 & -1 & 0 & 1 \end{array}\right]}$$

$A_R$: resistors $A_S$: sources

# $Y_n$ from the graph: the general result

Resistors first, then current sources: $A = [\, A_R \;\; A_S \,]$

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$$Y_n = A_R\, G\, A_R^T, \qquad \mathbf{i}_n = -A_S \mathbf{i}_S$$

$Y_n^T = Y_n$: always symmetric

# $Y_n$ from the graph: Example 11.1

From $A$: $A_R$ = columns 1–5, $A_S$ = columns 6, 7

$$A_R\, G = \begin{bmatrix} G_1 & G_2 & 0 & 0 & G_5 \\ 0 & -G_2 & G_3 & G_4 & 0 \\ 0 & 0 & 0 & -G_4 & -G_5 \end{bmatrix}$$

# $Y_n$ from the graph: Example 11.1

From $A$: $A_R$ = columns 1–5, $A_S$ = columns 6, 7

$$A_R\, G = \begin{bmatrix} G_1 & G_2 & 0 & 0 & G_5 \\ 0 & -G_2 & G_3 & G_4 & 0 \\ 0 & 0 & 0 & -G_4 & -G_5 \end{bmatrix}$$

$$A_R\, G\, A_R^T = \begin{bmatrix} G_1 + G_2 + G_5 & -G_2 & -G_5 \\ -G_2 & G_2 + G_3 + G_4 & -G_4 \\ -G_5 & -G_4 & G_4 + G_5 \end{bmatrix}$$

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$$-A_S \mathbf{i}_S = -\begin{bmatrix} -1 & 0 \\ 0 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} i_{s1} \\ i_{s2} \end{bmatrix} = \begin{bmatrix} i_{s1} \\ 0 \\ -i_{s2} \end{bmatrix} = \mathbf{i}_n \quad \checkmark$$

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The rules = this product

# Roadmap









**Current sources** | **Voltage sources** | **Dependent sources** | **The ground**

✓ | 2 | 3 | 4

the procedure, on an example ✓

matrix form ✓

**one end at the ground**

between two nodes

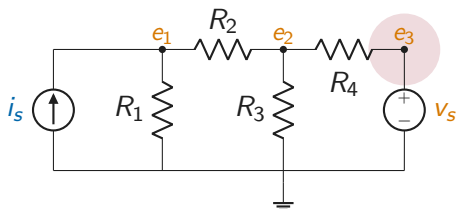
a dependent current source

a dependent voltage source

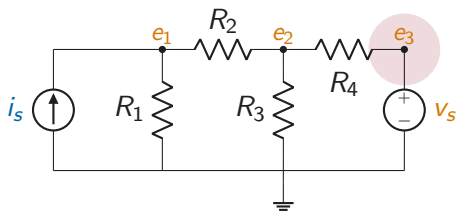
does its choice matter?

# Example 11.2: a voltage source to the ground

$e_3 = v_s$: known



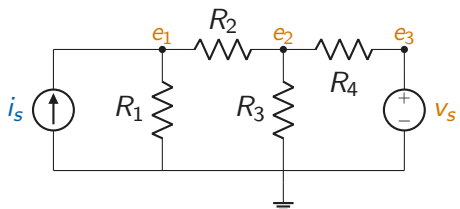
# Example 11.2: a voltage source to the ground




$e_3 = v_s$: known

No KCL at node 3. Unknowns: $e_1$, $e_2$

# Example 11.2: a voltage source to the ground

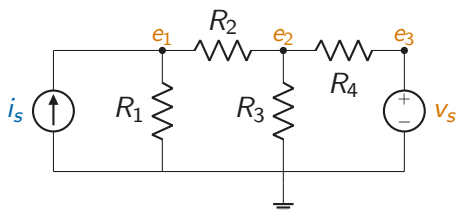



$e_3 = v_s$: known

No KCL at node 3. Unknowns: $e_1$, $e_2$

**Node 1:** $-i_s + \dfrac{e_1}{R_1} + \dfrac{e_1 - e_2}{R_2} = 0$

# Example 11.2: a voltage source to the ground



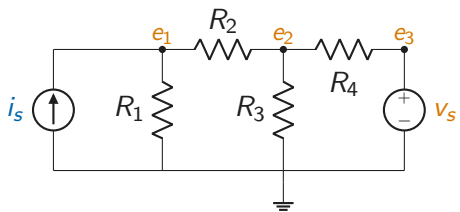

$e_3 = v_s$: known

No KCL at node 3. Unknowns: $e_1$, $e_2$

**Node 1:** $$-i_s + \frac{e_1}{R_1} + \frac{e_1 - e_2}{R_2} = 0$$

**Node 2:** $$\frac{e_2 - e_1}{R_2} + \frac{e_2}{R_3} + \frac{e_2 - v_s}{R_4} = 0$$

# Example 11.2: a voltage source to the ground




$e_3 = v_s$: known

No KCL at node 3. Unknowns: $e_1$, $e_2$

**Node 1:** $-i_s + \dfrac{e_1}{R_1} + \dfrac{e_1 - e_2}{R_2} = 0$

**Node 2:** $\dfrac{e_2 - e_1}{R_2} + \dfrac{e_2}{R_3} + \dfrac{e_2 - v_s}{R_4} = 0$

$$\begin{bmatrix} \frac{1}{R_1} + \frac{1}{R_2} & -\frac{1}{R_2} \\ -\frac{1}{R_2} & \frac{1}{R_2} + \frac{1}{R_3} + \frac{1}{R_4} \end{bmatrix} \begin{bmatrix} e_1 \\ e_2 \end{bmatrix} = \begin{bmatrix} i_s \\ v_s/R_4 \end{bmatrix}$$

# Roadmap









**Current sources** | **Voltage sources** | **Dependent sources** | **The ground**

✓ | 2 | 3 | 4

the procedure, on an example ✓

matrix form ✓

one end at the ground ✓

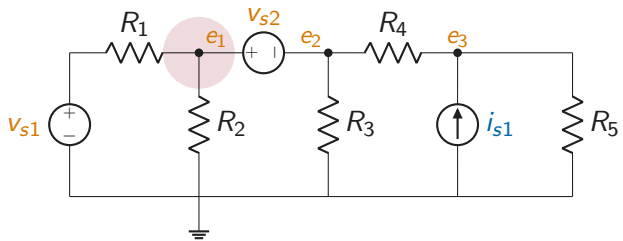
**between two nodes**

a dependent current source

a dependent voltage source

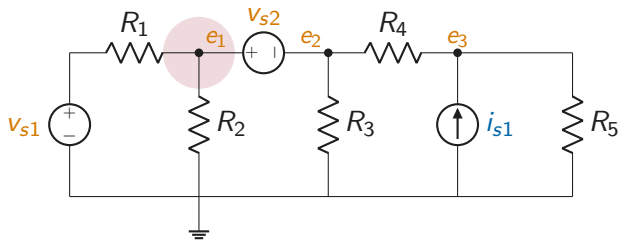
does its choice matter?

# Example 11.3: a voltage source between two nodes



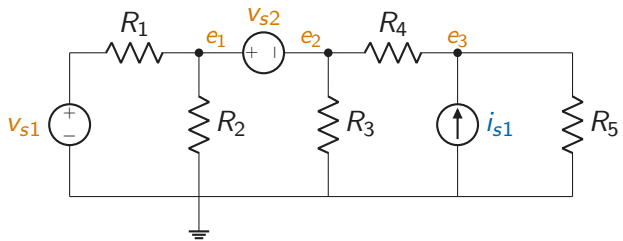

- **Question.** KCL at node 1: current through $v_{s2}$?

# Example 11.3: a voltage source between two nodes



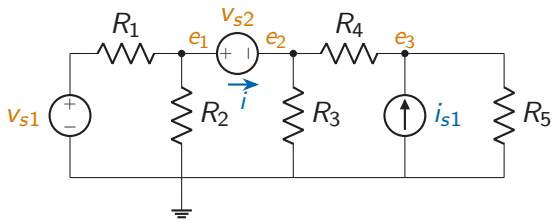

- **Question.** KCL at node 1: current through $v_{s2}$?
- Not given by $e_1$, $e_2$: a voltage source fixes $v$, not $i$

# Example 11.3: a voltage source between two nodes



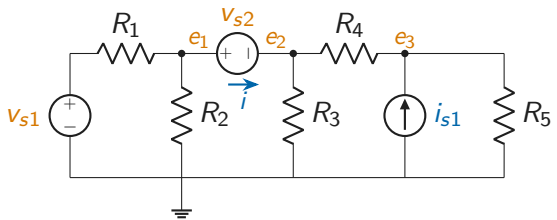

- **Question.** KCL at node 1: current through $v_{s2}$?
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- Method I: its current as an unknown  Method II: supernode

# Method I: the current as an unknown




New unknown: $i$

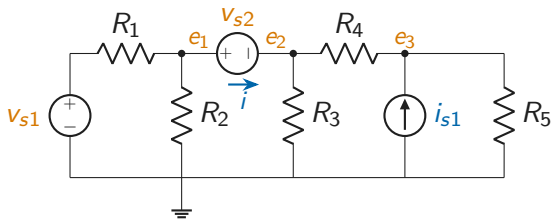
# Method I: the current as an unknown




**Node 1:** $\dfrac{e_1 - v_{s1}}{R_1} + \dfrac{e_1}{R_2} + i = 0$

New unknown: $i$

# Method I: the current as an unknown

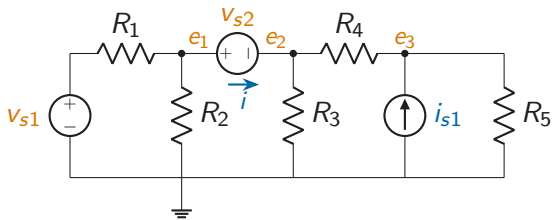



**Node 1:** $\dfrac{e_1 - v_{s1}}{R_1} + \dfrac{e_1}{R_2} + i = 0$

New unknown: $i$

**Node 2:** $-i + \dfrac{e_2}{R_3} + \dfrac{e_2 - e_3}{R_4} = 0$

# Method I: the current as an unknown



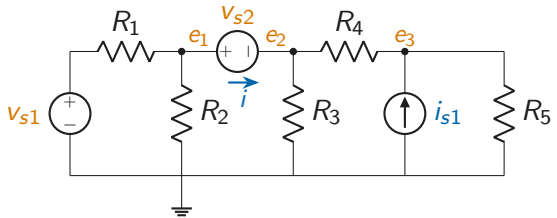

**Node 1:** $\dfrac{e_1 - v_{s1}}{R_1} + \dfrac{e_1}{R_2} + i = 0$

**Node 2:** $-i + \dfrac{e_2}{R_3} + \dfrac{e_2 - e_3}{R_4} = 0$

**Node 3:** $\dfrac{e_3 - e_2}{R_4} - i_{s1} + \dfrac{e_3}{R_5} = 0$

New unknown: $i$

# Method I: the current as an unknown




**Node 1:** $\dfrac{e_1 - v_{s1}}{R_1} + \dfrac{e_1}{R_2} + i = 0$

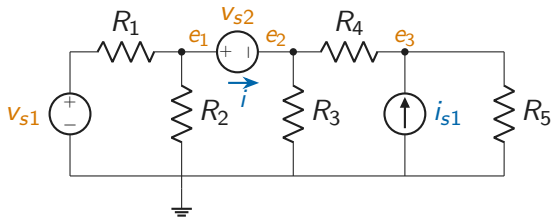
**Node 2:** $-i + \dfrac{e_2}{R_3} + \dfrac{e_2 - e_3}{R_4} = 0$

**Node 3:** $\dfrac{e_3 - e_2}{R_4} - i_{s1} + \dfrac{e_3}{R_5} = 0$

New unknown: $i$

**Constraint:** $e_1 - e_2 = v_{s2}$

# Method I: the current as an unknown



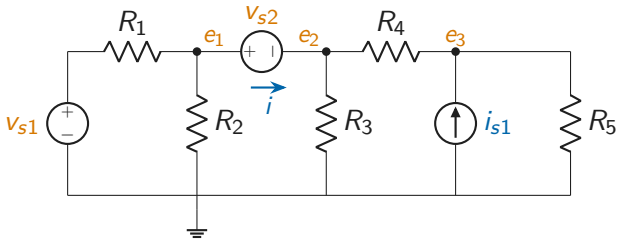

$$\begin{bmatrix} G_1+G_2 & 0 & 0 & 1 \\ 0 & G_3+G_4 & -G_4 & -1 \\ 0 & -G_4 & G_4+G_5 & 0 \\ 1 & -1 & 0 & 0 \end{bmatrix} \begin{bmatrix} e_1 \\ e_2 \\ e_3 \\ i \end{bmatrix} = \begin{bmatrix} G_1 v_{s1} \\ 0 \\ i_{s1} \\ v_{s2} \end{bmatrix}$$

New unknown: $i$

**Constraint:** $e_1 - e_2 = v_{s2}$

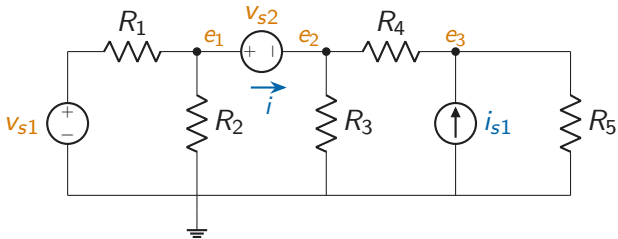
4 unknowns, 4 equations

# Method II: the supernode




**Question.** Can we get rid of $i$?

# Method II: the supernode

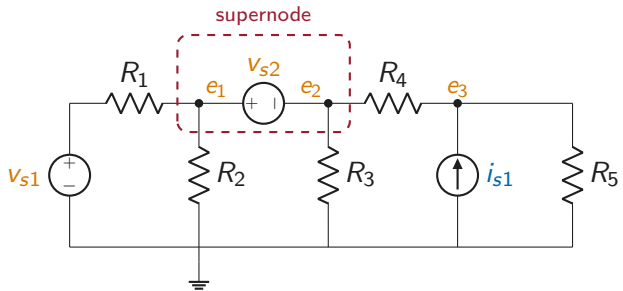



**Question.** Can we get rid of $i$?

Add KCL of nodes 1 and 2:

$i$ cancels

# Method II: the supernode

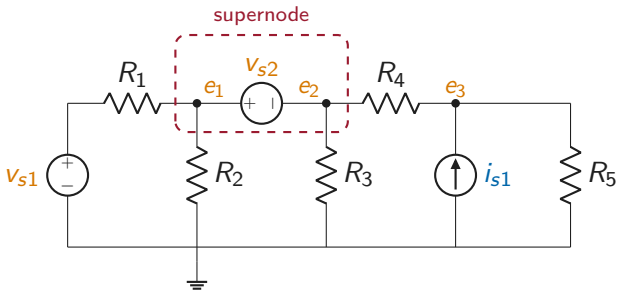



**Question.** Can we get rid of $i$? = KCL on a closed surface

Add KCL of nodes 1 and 2:

$i$ cancels

# Method II: the supernode




**Question.** Can we get rid of $i$?

Add KCL of nodes 1 and 2:

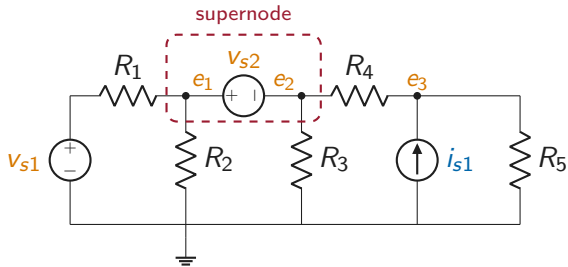
$i$ cancels

= KCL on a closed surface

**Supernode**

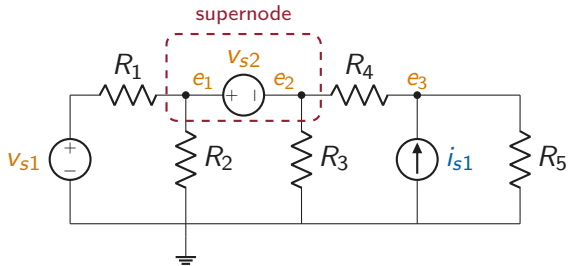
Two nodes of the source as one node

# Method II: the equations




**Question.** KCL at the supernode?

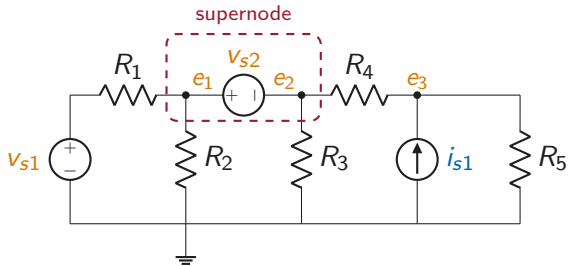
# Method II: the equations




**Supernode:**

$$\frac{e_1 - v_{s1}}{R_1} + \frac{e_1}{R_2} + \frac{e_2}{R_3} + \frac{e_2 - e_3}{R_4} = 0$$

# Method II: the equations

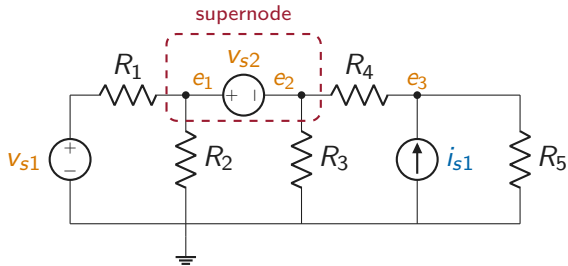



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**Node 3:** $\dfrac{e_3 - e_2}{R_4} - i_{s1} + \dfrac{e_3}{R_5} = 0$

# Method II: the equations



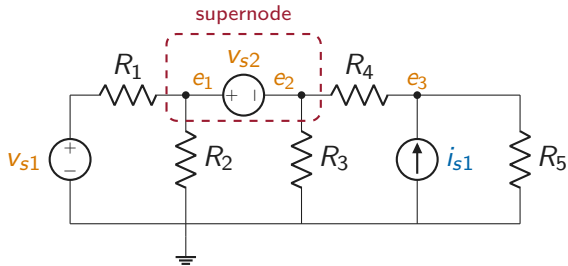

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**Node 3:** $\dfrac{e_3 - e_2}{R_4} - i_{s1} + \dfrac{e_3}{R_5} = 0$

**Constraint:** $e_1 - e_2 = v_{s2}$

# Method II: the equations




**Supernode:**

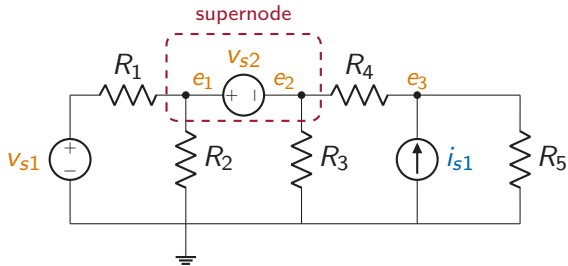
$$\frac{e_1 - v_{s1}}{R_1} + \frac{e_1}{R_2} + \frac{e_2}{R_3} + \frac{e_2 - e_3}{R_4} = 0$$

**Node 3:** $\dfrac{e_3 - e_2}{R_4} - i_{s1} + \dfrac{e_3}{R_5} = 0$

**Constraint:** $e_1 - e_2 = v_{s2}$

$$\begin{bmatrix} G_1 + G_2 & G_3 + G_4 & -G_4 \\ 0 & -G_4 & G_4 + G_5 \\ 1 & -1 & 0 \end{bmatrix} \begin{bmatrix} e_1 \\ e_2 \\ e_3 \end{bmatrix} = \begin{bmatrix} G_1 v_{s1} \\ i_{s1} \\ v_{s2} \end{bmatrix}$$

# Method II: the equations




**Supernode:**

$$\frac{e_1 - v_{s1}}{R_1} + \frac{e_1}{R_2} + \frac{e_2}{R_3} + \frac{e_2 - e_3}{R_4} = 0$$

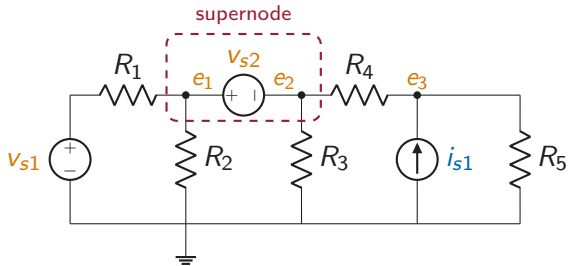
**Node 3:** $\dfrac{e_3 - e_2}{R_4} - i_{s1} + \dfrac{e_3}{R_5} = 0$

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3 unknowns (one fewer)

# Method II: the equations




**Supernode:**

$$\frac{e_1 - v_{s1}}{R_1} + \frac{e_1}{R_2} + \frac{e_2}{R_3} + \frac{e_2 - e_3}{R_4} = 0$$

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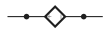
3 unknowns (one fewer)

Not symmetric

# Roadmap









**Current sources**

✓ the procedure, on an example ✓

matrix form ✓

**Voltage sources**

✓ one end at the ground ✓

between two nodes ✓

**Dependent sources**

3

**a dependent current source**

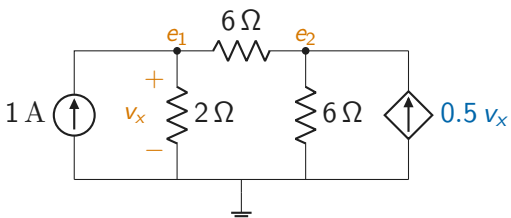
**a dependent voltage source**

**The ground**

4

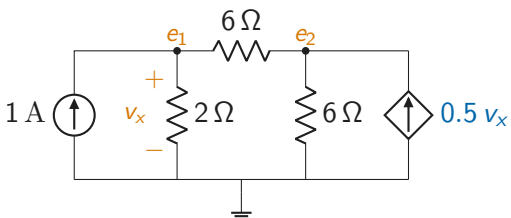
does its choice matter?

# Example 11.4: a dependent current source




**Question.** Find $e_1$ and $e_2$.

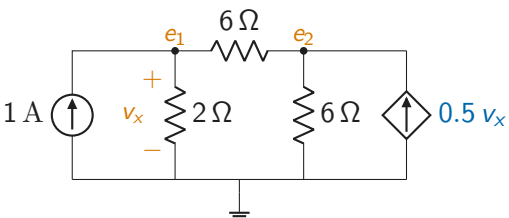
# Example 11.4: a dependent current source




**Question.** Find $e_1$ and $e_2$.

Treat $0.5\,v_x$ as a current source

# Example 11.4: a dependent current source

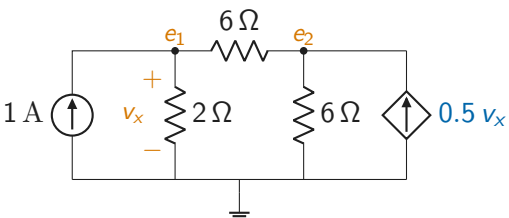



**Question.** Find $e_1$ and $e_2$.

Treat $0.5\,v_x$ as a current source

**Node 1:** $-1 + \dfrac{e_1}{2} + \dfrac{e_1 - e_2}{6} = 0$

# Example 11.4: a dependent current source



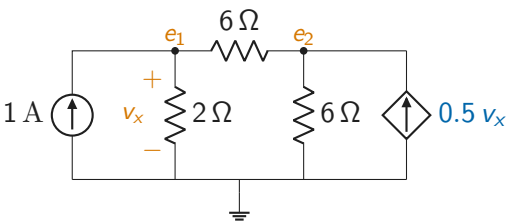

**Question.** Find $e_1$ and $e_2$.

Treat $0.5\,v_x$ as a current source

**Node 1:** $$-1 + \frac{e_1}{2} + \frac{e_1 - e_2}{6} = 0$$

**Node 2:** $$\frac{e_2 - e_1}{6} + \frac{e_2}{6} - 0.5\,v_x = 0$$

# Example 11.4: a dependent current source




**Question.** Find $e_1$ and $e_2$.

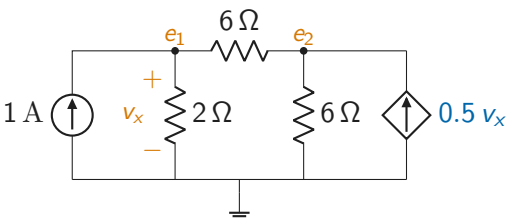
Treat $0.5\,v_x$ as a current source

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**Question.** $v_x$ in node voltages?

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**Question.** $v_x$ in node voltages?

$v_x = e_1$

# Example 11.4: matrix form

**First** plug $v_x = e_1$ into node 2:

$$\frac{e_2 - e_1}{6} + \frac{e_2}{6} - 0.5\, e_1 = 0$$

# Example 11.4: matrix form

**First** plug $v_x = e_1$ into node 2:

$$\frac{e_2 - e_1}{6} + \frac{e_2}{6} - 0.5\, e_1 = 0 \qquad \Rightarrow \qquad -\frac{2}{3}\, e_1 + \frac{1}{3}\, e_2 = 0$$

# Example 11.4: matrix form

**First** plug $v_x = e_1$ into node 2:

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$$\begin{bmatrix} \frac{2}{3} & -\frac{1}{6} \\ -\frac{2}{3} & \frac{1}{3} \end{bmatrix} \begin{bmatrix} e_1 \\ e_2 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

# Example 11.4: matrix form

**First** plug $v_x = e_1$ into node 2:

$$\frac{e_2 - e_1}{6} + \frac{e_2}{6} - 0.5\, e_1 = 0 \qquad \Rightarrow \qquad -\frac{2}{3}\, e_1 + \frac{1}{3}\, e_2 = 0$$

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# Example 11.4: matrix form

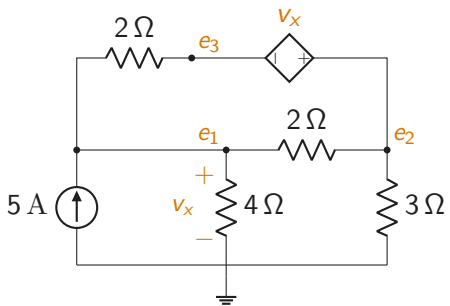
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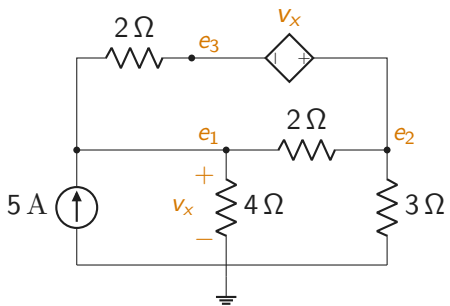
$-\frac{1}{6} \neq -\frac{2}{3}$: not symmetric

# Example 11.5: a dependent voltage source




**Question.** Power absorbed by the 3 Ω?

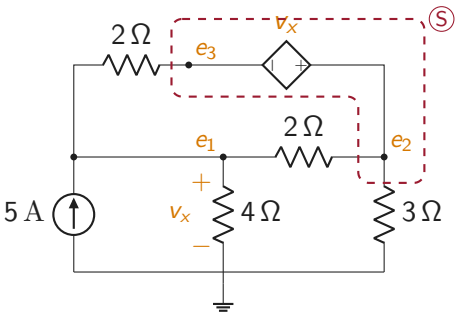
# Example 11.5: a dependent voltage source




**Question.** Power absorbed by the 3 Ω?

$v_x$: between nodes 3 and 2, no ground

# Example 11.5: a dependent voltage source

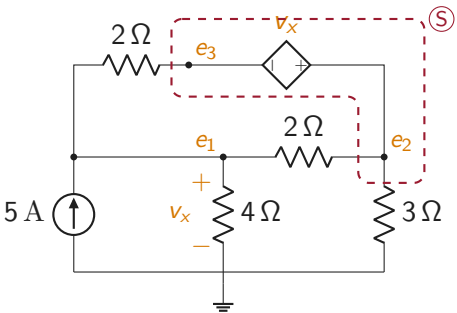



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$\Rightarrow$ supernode (S)

# Example 11.5: a dependent voltage source



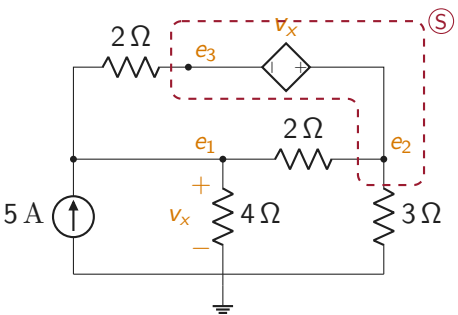

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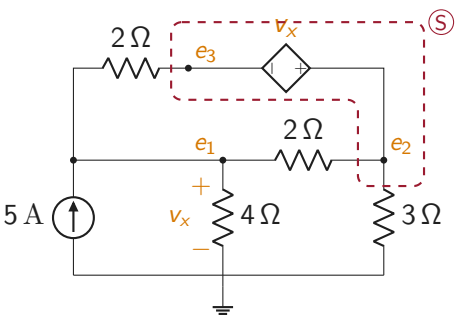
Unknowns: $e_1$, $e_2$, $e_3$

# Example 11.5: the equations




**Node 1:** $$-5 + \frac{e_1 - e_3}{2} + \frac{e_1}{4} + \frac{e_1 - e_2}{2} = 0 \quad (1)$$

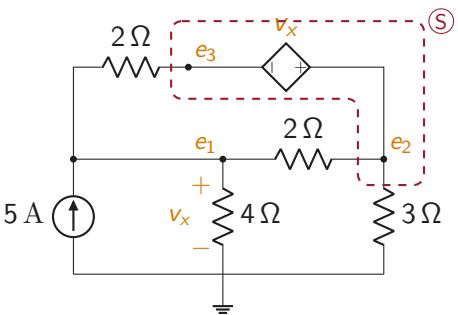
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**Node 1:** $$-5 + \frac{e_1 - e_3}{2} + \frac{e_1}{4} + \frac{e_1 - e_2}{2} = 0 \quad (1)$$

**Supernode Ⓢ:** $$\frac{e_3 - e_1}{2} + \frac{e_2 - e_1}{2} + \frac{e_2}{3} = 0 \quad (2)$$

# Example 11.5: the equations

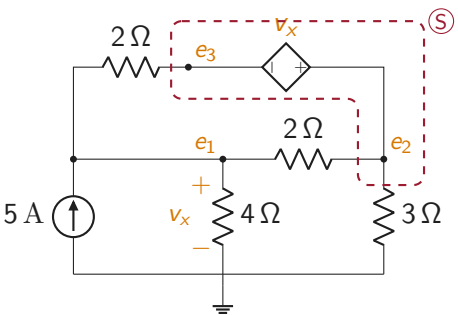



**Node 1:** $-5 + \dfrac{e_1 - e_3}{2} + \dfrac{e_1}{4} + \dfrac{e_1 - e_2}{2} = 0 \quad (1)$

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**Question.** The constraint?

# Example 11.5: the equations



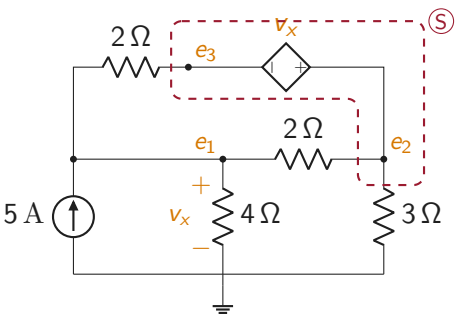

**Node 1:** $-5 + \dfrac{e_1 - e_3}{2} + \dfrac{e_1}{4} + \dfrac{e_1 - e_2}{2} = 0$ (1)

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**Question.** The constraint?

Source: $e_2 - e_3 = v_x$

# Example 11.5: the equations




**Node 1:** $-5 + \dfrac{e_1 - e_3}{2} + \dfrac{e_1}{4} + \dfrac{e_1 - e_2}{2} = 0$ (1)

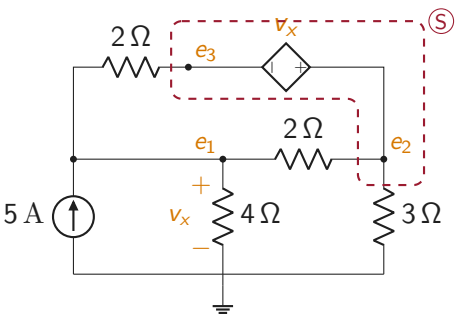
**Supernode Ⓢ:** $\dfrac{e_3 - e_1}{2} + \dfrac{e_2 - e_1}{2} + \dfrac{e_2}{3} = 0$ (2)

**Question.** The constraint?

Source: $e_2 - e_3 = v_x$

Control: $v_x = e_1$

# Example 11.5: the equations




**Node 1:** $-5 + \dfrac{e_1 - e_3}{2} + \dfrac{e_1}{4} + \dfrac{e_1 - e_2}{2} = 0$ (1)

**Supernode Ⓢ:** $\dfrac{e_3 - e_1}{2} + \dfrac{e_2 - e_1}{2} + \dfrac{e_2}{3} = 0$ (2)

**Question.** The constraint?

Source: $e_2 - e_3 = v_x$

Control: $v_x = e_1$

**Constraint:** $e_2 - e_3 = e_1$ (3)

# Example 11.5: matrix form

(1), (2), (3):

$$\begin{bmatrix} \frac{5}{4} & -\frac{1}{2} & -\frac{1}{2} \\ -1 & \frac{5}{6} & \frac{1}{2} \\ 1 & -1 & 1 \end{bmatrix} \begin{bmatrix} e_1 \\ e_2 \\ e_3 \end{bmatrix} = \begin{bmatrix} 5 \\ 0 \\ 0 \end{bmatrix}$$

# Example 11.5: matrix form

(1), (2), (3):

$$\begin{bmatrix} \frac{5}{4} & -\frac{1}{2} & -\frac{1}{2} \\ -1 & \frac{5}{6} & \frac{1}{2} \\ 1 & -1 & 1 \end{bmatrix} \begin{bmatrix} e_1 \\ e_2 \\ e_3 \end{bmatrix} = \begin{bmatrix} 5 \\ 0 \\ 0 \end{bmatrix}$$

$$e_1 = 8\,\mathrm{V}, \quad e_2 = 9\,\mathrm{V}, \quad e_3 = 1\,\mathrm{V}$$

# Example 11.5: matrix form

(1), (2), (3):

$$\begin{bmatrix} \frac{5}{4} & -\frac{1}{2} & -\frac{1}{2} \\ -1 & \frac{5}{6} & \frac{1}{2} \\ 1 & -1 & 1 \end{bmatrix} \begin{bmatrix} e_1 \\ e_2 \\ e_3 \end{bmatrix} = \begin{bmatrix} 5 \\ 0 \\ 0 \end{bmatrix}$$

$$e_1 = 8\,\mathrm{V}, \quad e_2 = 9\,\mathrm{V}, \quad e_3 = 1\,\mathrm{V}$$

$$P_{3\Omega} = \frac{{e_2}^2}{3} = \frac{9^2}{3} = 27\,\mathrm{W}$$

# Dependent sources

- Like an independent source of the same kind

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- Control variable in node voltages

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- First plug it in, then form the matrix

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- Like an independent source of the same kind
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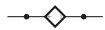
**Remark**

Dependent voltage source from node $k$ to the ground:
no KCL at node $k$; write the constraint instead

# Roadmap









**Current sources**

- the procedure, on an example ✓
- matrix form ✓

**Voltage sources**

- one end at the ground ✓
- between two nodes ✓

**Dependent sources**

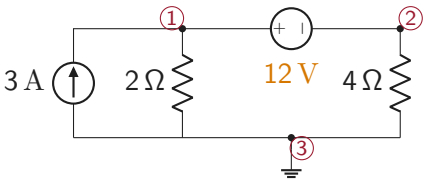
- a dependent current source ✓
- a dependent voltage source ✓

**The ground**

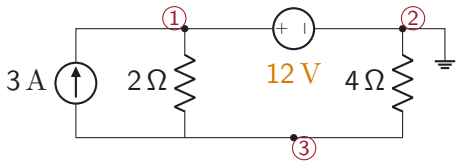
4

**does its choice matter?**

# Example 11.6: where to put the ground?



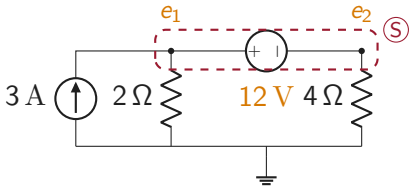

Ground at node 3

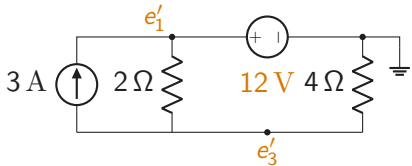



Ground at node 2: the − terminal of 12 V

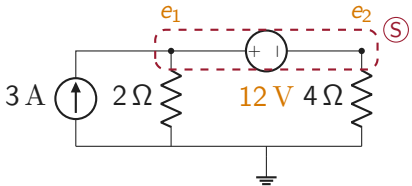
**Question.** Solve both. Which is easier?

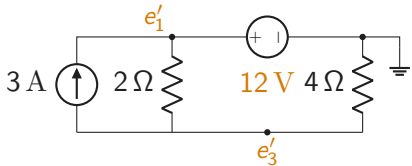
# Example 11.6: side by side



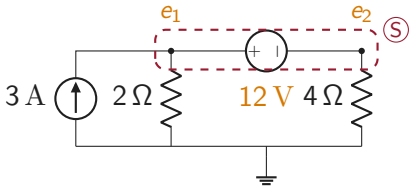
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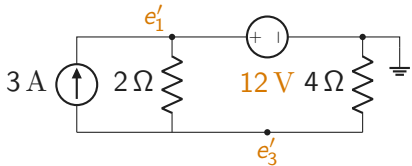



12 V between two nodes: supernode

# Example 11.6: side by side

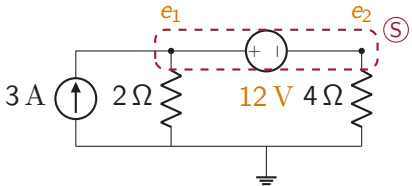


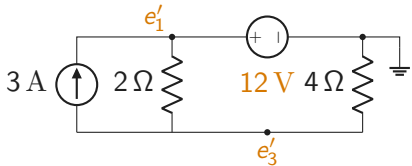




12 V between two nodes: supernode

**S:** $-3 + \frac{e_1}{2} + \frac{e_2}{4} = 0$

# Example 11.6: side by side



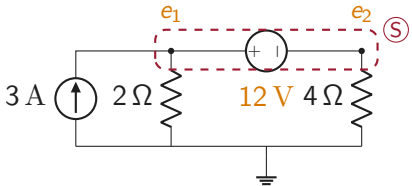



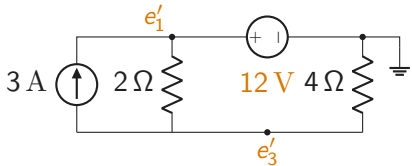

12 V between two nodes: supernode

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**Constraint:** $e_1 - e_2 = 12$

# Example 11.6: side by side



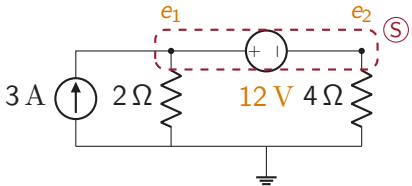

12 V between two nodes: supernode

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**Constraint:** $e_1 - e_2 = 12$

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# Example 11.6: side by side

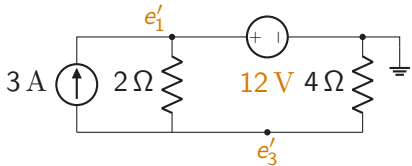



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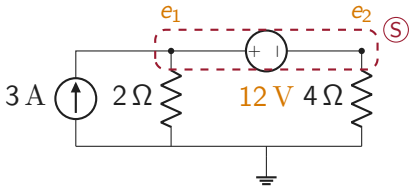
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$e_1' = 12\,\text{V}$: known

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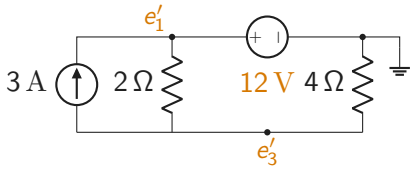



12 V between two nodes: supernode

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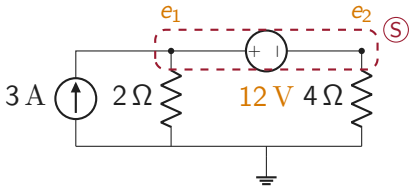
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**Node 3:** $3 + \frac{e_3' - 12}{2} + \frac{e_3'}{4} = 0$

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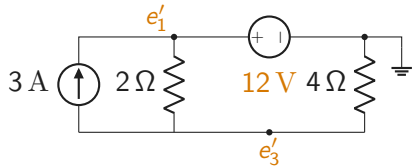



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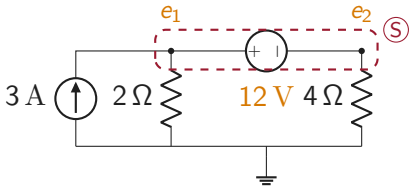



$e_1' = 12\,\text{V}$: known

**Node 3:** $3 + \frac{e_3' - 12}{2} + \frac{e_3'}{4} = 0$

$\frac{3}{4}\, e_3' = 3 \;\Rightarrow\; e_3' = 4\,\text{V}$

# Example 11.6: side by side

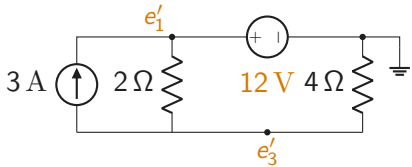



12 V between two nodes: supernode

**S:** $-3 + \frac{e_1}{2} + \frac{e_2}{4} = 0$

**Constraint:** $e_1 - e_2 = 12$

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2 unknowns | 1 unknown

# Example 11.6: the same branch voltages

| | ground at 3 | ground at 2 |
|---|---|---|
| $v_{2\Omega}$: node 1 to 3 | $8 - 0 = 8$ | $12 - 4 = 8$ |
| $v_{4\Omega}$: node 2 to 3 | $-4 - 0 = -4$ | $0 - 4 = -4$ |

(V)

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(V)

Node voltages shift by $4\,\mathrm{V}$; branch voltages and currents do not change

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(V)

Node voltages shift by $4\,\mathrm{V}$; branch voltages and currents do not change

**Tip.** Put the ground at the $-$ terminal of a voltage source: one node voltage is known, and no supernode is needed

# Summary

| In the circuit | What to do |
|---|---|
| resistors, current sources | $Y_n\mathbf{e} = \mathbf{i}_n$, by the rules; symmetric |
| voltage source to the ground | $e_k$ known; no KCL at node $k$ |
| voltage source between nodes | Method I or supernode, plus constraint |
| dependent source | control in node voltages: constraint |
| choice of ground | any node; best at the $-$ terminal of a voltage source |

Then $\mathbf{v} = A^T\mathbf{e}$ and the element laws